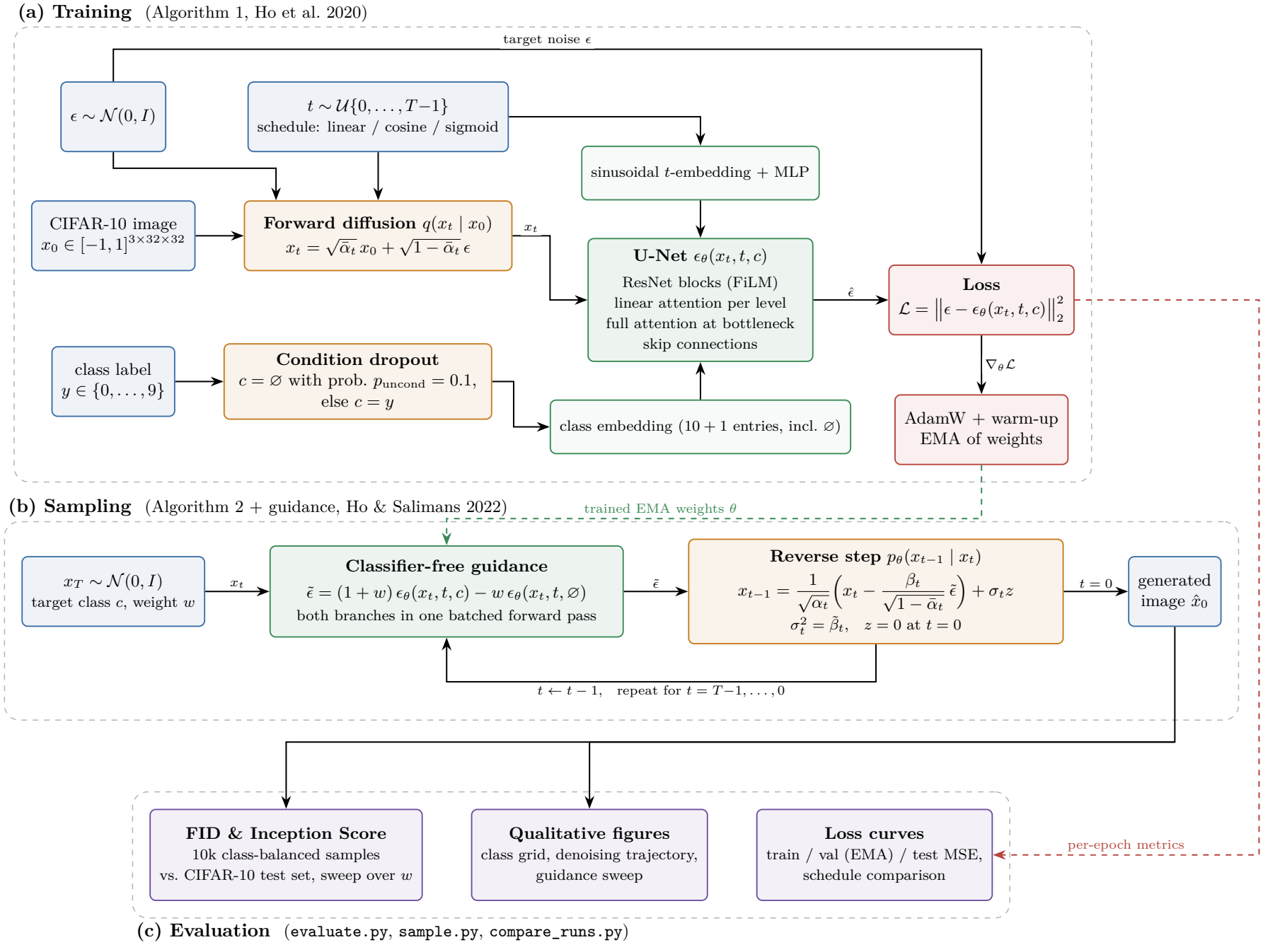

(a) Training (Algorithm 1, Ho et al. 2020)
target noise ϵ
$\epsilon \sim \mathcal{N}(0, I)$
$t \sim \mathcal{U}\{0, \ldots, T-1\}$
schedule: linear / cosine / sigmoid
sinusoidal t-embedding + MLP
CIFAR-10 image
$x_0 \in [-1, 1]^{3\times32\times32}$
Forward diffusion $q(x_t \mid x_0)$
$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$
x_t
U-Net $\epsilon_\theta(x_t, t, c)$
ResNet blocks (FiLM)
linear attention per level
full attention at bottleneck
skip connections
$\hat{\epsilon}$
Loss
$\mathcal{L} = \left\|\epsilon - \epsilon_\theta(x_t, t, c)\right\|_2^2$
$\nabla_\theta \mathcal{L}$
class label
$y \in \{0, \ldots, 9\}$
Condition dropout
$c = \varnothing$ with prob. $p_{\text{uncond}} = 0.1$,
else $c = y$
class embedding ($10 + 1$ entries, incl. $\varnothing$)
AdamW + warm-up
EMA of weights
(b) Sampling (Algorithm 2 + guidance, Ho & Salimans 2022)
trained EMA weights θ
$x_T \sim \mathcal{N}(0, I)$
target class c, weight w
x_t
Classifier-free guidance
$\tilde{\epsilon} = (1 + w)\, \epsilon_\theta(x_t, t, c) - w\, \epsilon_\theta(x_t, t, \varnothing)$
both branches in one batched forward pass
$\tilde{\epsilon}$
Reverse step $p_\theta(x_{t-1} \mid x_t)$
$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\tilde{\epsilon}\right) + \sigma_t z$
$\sigma_t^2 = \tilde{\beta}_t, \quad z = 0$ at $t = 0$
$t = 0$
generated image $\hat{x}_0$
$t \leftarrow t - 1$, repeat for $t = T-1, \ldots, 0$
FID & Inception Score
10k class-balanced samples
vs. CIFAR-10 test set, sweep over w
Qualitative figures
class grid, denoising trajectory,
guidance sweep
Loss curves
train / val (EMA) / test MSE,
schedule comparison
per-epoch metrics
(c) Evaluation (evaluate.py, sample.py, compare_runs.py)